



AKSHAYKALA AI STUDIO

Generative AI Engine for Fashion
Mockup Synthesis

Introduction

Background

Jewelry is one of the highest-consideration purchases in e-commerce. A customer cannot feel the weight or hold it up to their ear. The product image is the product. For a brand releasing hundreds of SKUs per season, every single one needs a clean product shot, a lifestyle shot, and an on-model shot. That visual pipeline is the difference between a listing that converts and one that does not.

Problem Statement

The industry standard for producing those images is a traditional studio photoshoot with models, lighting rigs, photographers, post-production editors. For a brand like AkshayKala, this means thousands of rupees per session, days of turnaround time per batch, and a pipeline that simply cannot keep up with catalogue growth. At 50 SKUs it is manageable. At 500, it is a crisis. Cost scales linearly, speed does not improve, and visual consistency across sessions is impossible to guarantee.

Solution

An automated, containerized Computer Vision and Generative AI pipeline.



AkshayKala AI Studio - Earring Mockup Engine

Professional automated batch processing for Studio, Abstract, and On-Model jewelry mockups.

Input Settings

Upload Earring ZIP

Gradio_demo_test.zip329.9 KB

Abstract Background Template (Optional)

abstract_background.jpg194.1 KB

Fashion Model Template (Optional)

Drop File Here

- or -

Click to Upload

Output Variations

☒ White Background

☒ Abstract Background

☒ Model Mockup (Uses API Credits)

Generate Mockups

Status

Results Gallery

Generated Variations



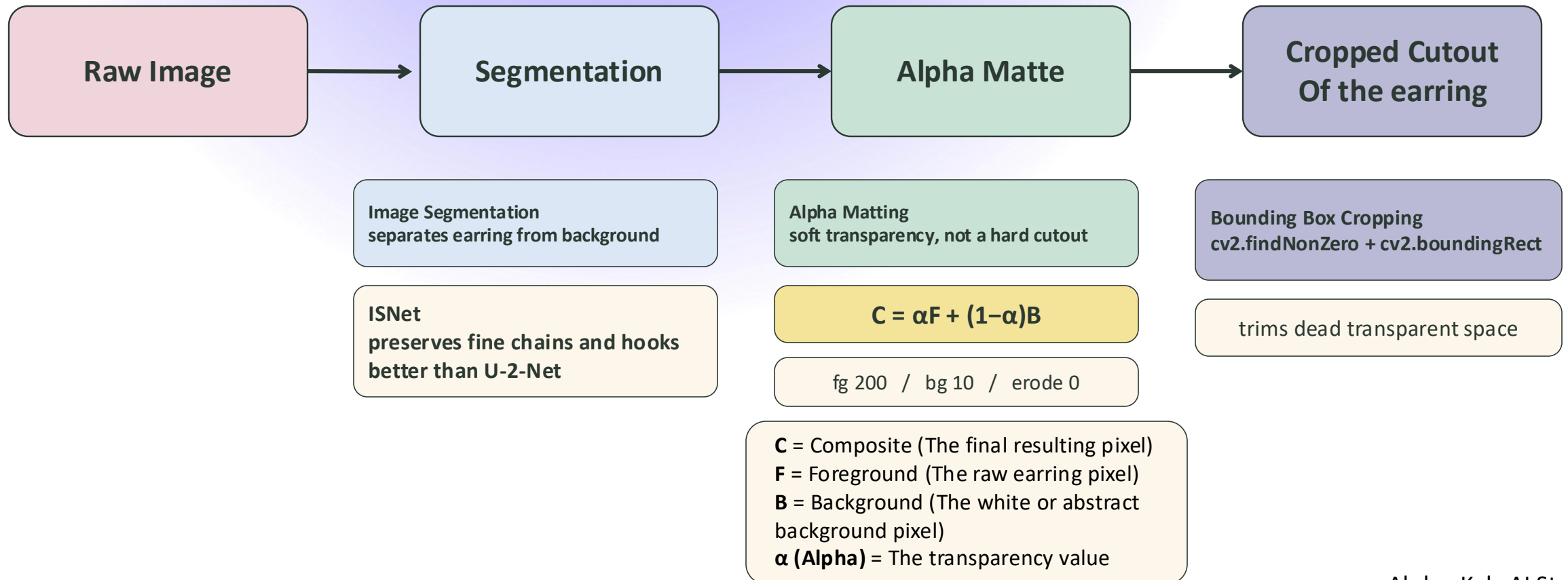


Download Full Batch (ZIP)

Final_Mockups.zip2.2 MB

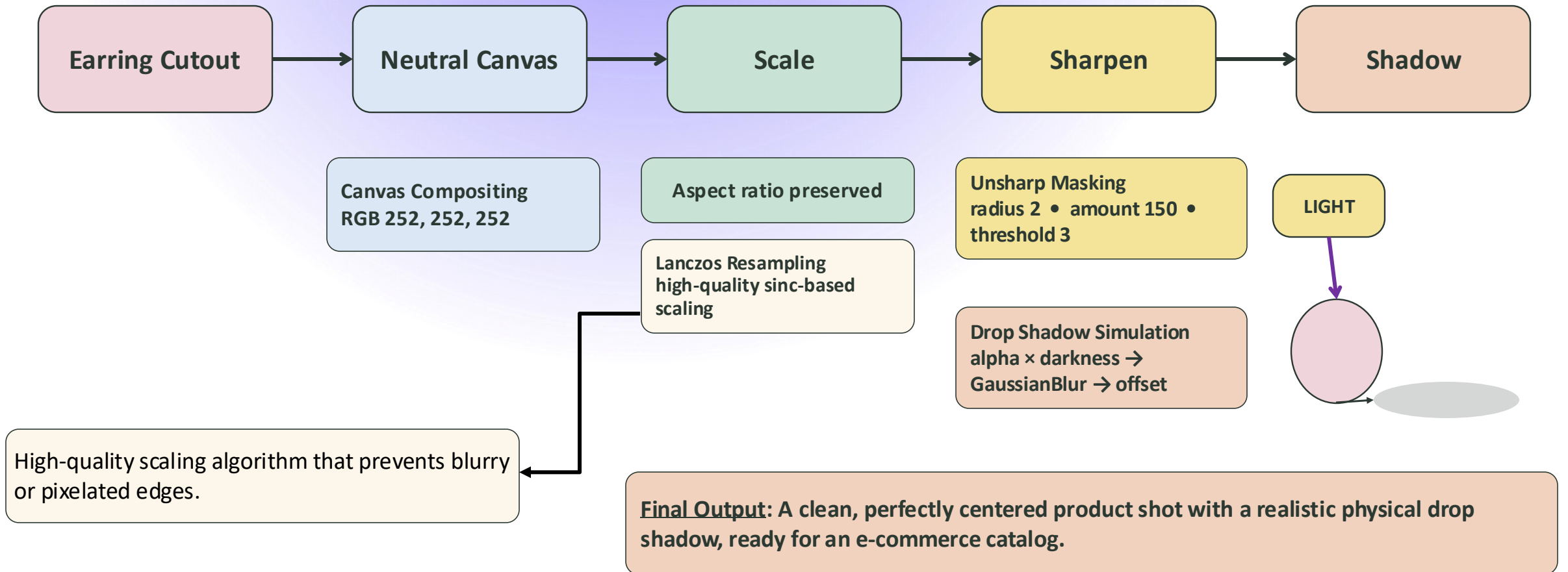
The Technical Framework

Computer Vision Pipeline (Common to all 3 outputs):



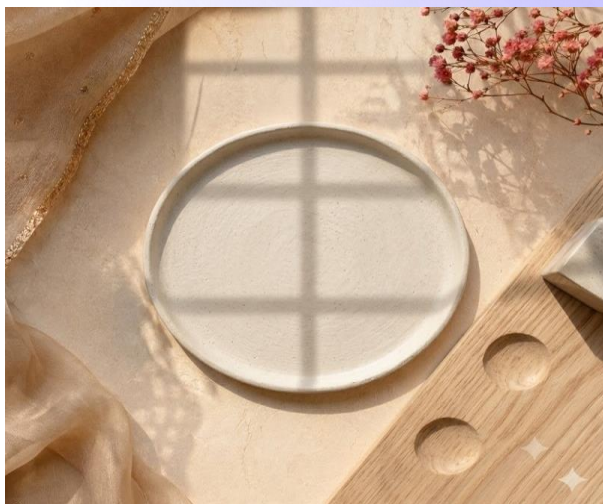
Output 1: White Studio Background

A controlled product shot, built from the shared earring cutout



Output 2: Find the Placement Surface

Auto Plate Detection isolates a stable compositing region



BACKGROUND PHOTO



CANNY EDGES + DILATE

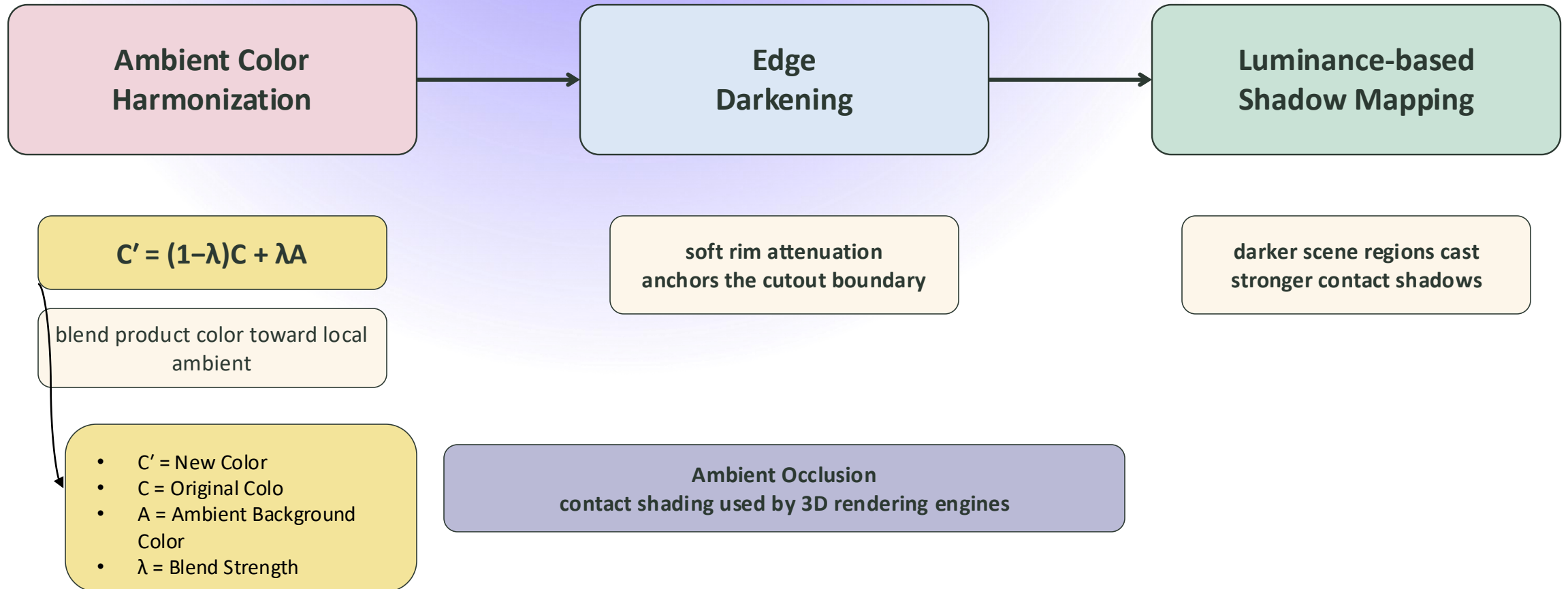


FIND CONTOURS

Contour Area Filtering
rejects noise, keeps the dominant plate

Output 2: Make the Composite Belong

Color and light cues bind the earring to the scene



Output 3: On-model Mockup

Two Strategies: One production track, one long-term R&D track

STRATEGY A / PRODUCTION

Local Diffusion

Stable Diffusion foundation

LoRA + DreamBooth adaptation

Inpainting + ControlNet conditioning

- control
- self-hosting
- no credit cost

STRATEGY B / PRODUCTION

Generative API

Claid.ai MLaaS endpoint

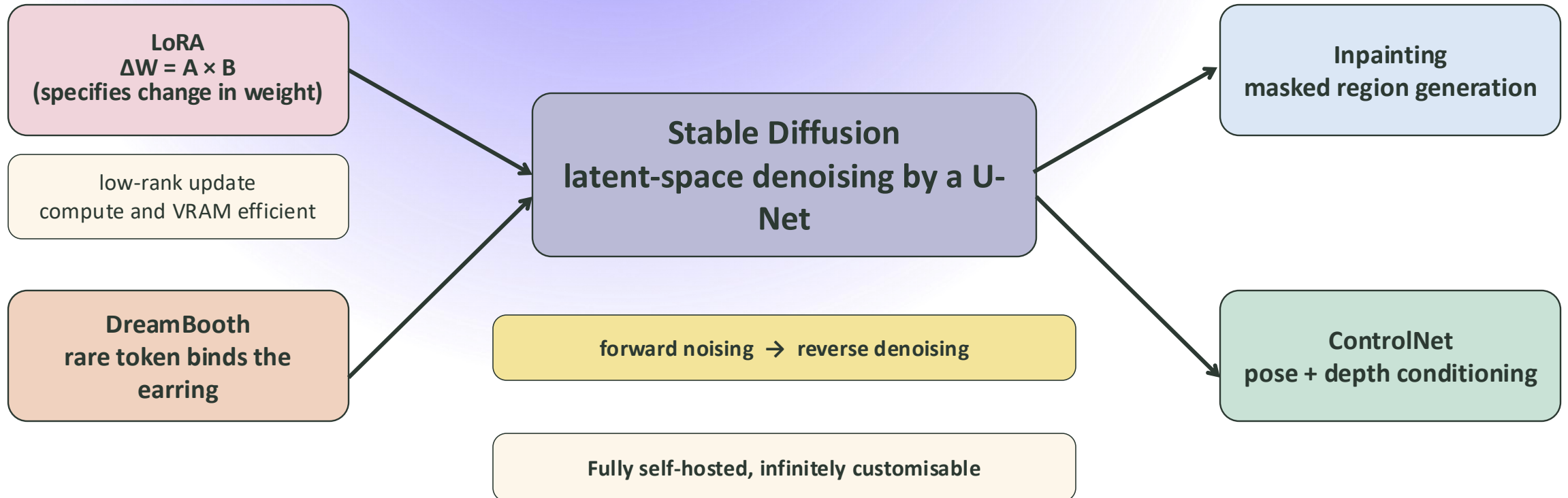
CDN → REST job → polling → result

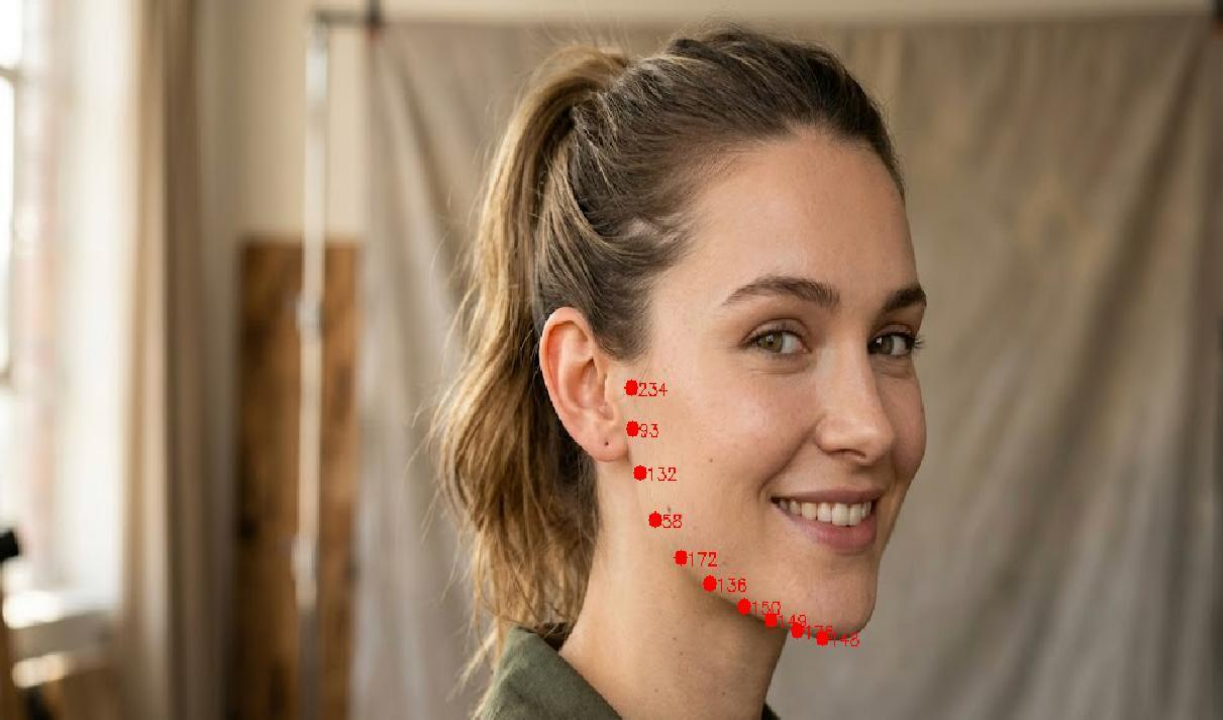
Constrained prompt guidance

- speed to ship
- consistent output

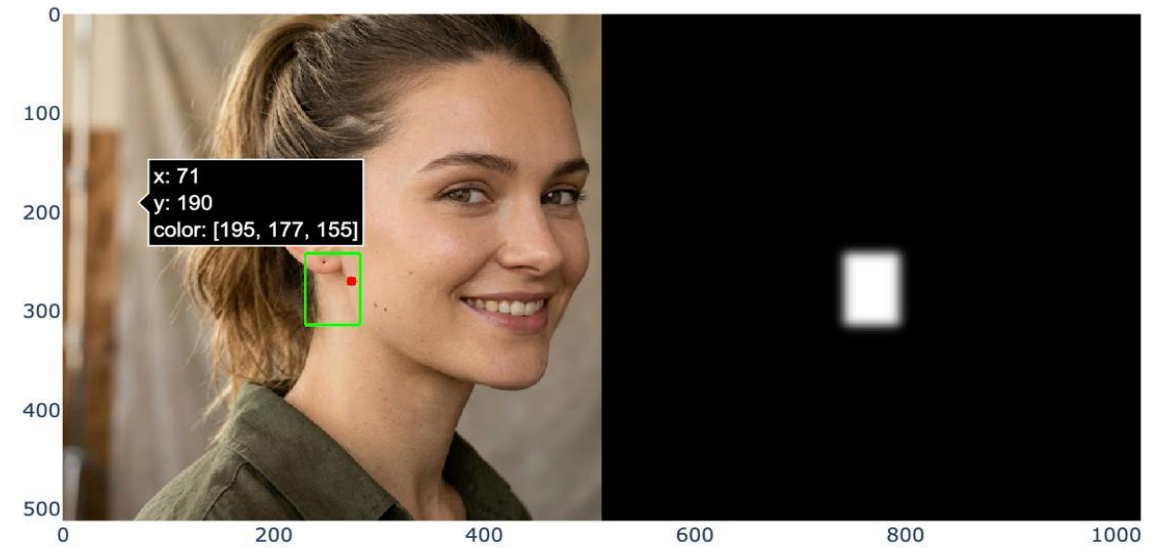
Strategy A: Local Diffusion

Adapt the subject, constrain the region, preserve the pose





512x512 Crop: Image vs. Mask



MediaPipe_Face_Mesh Draft saved

File Edit View Run Settings Add-ons Help

Code

```
fig = px.imshow(np.hstack([mask, feathered_mask]), binary_string=True, title="Raw Mask vs. Feath
```

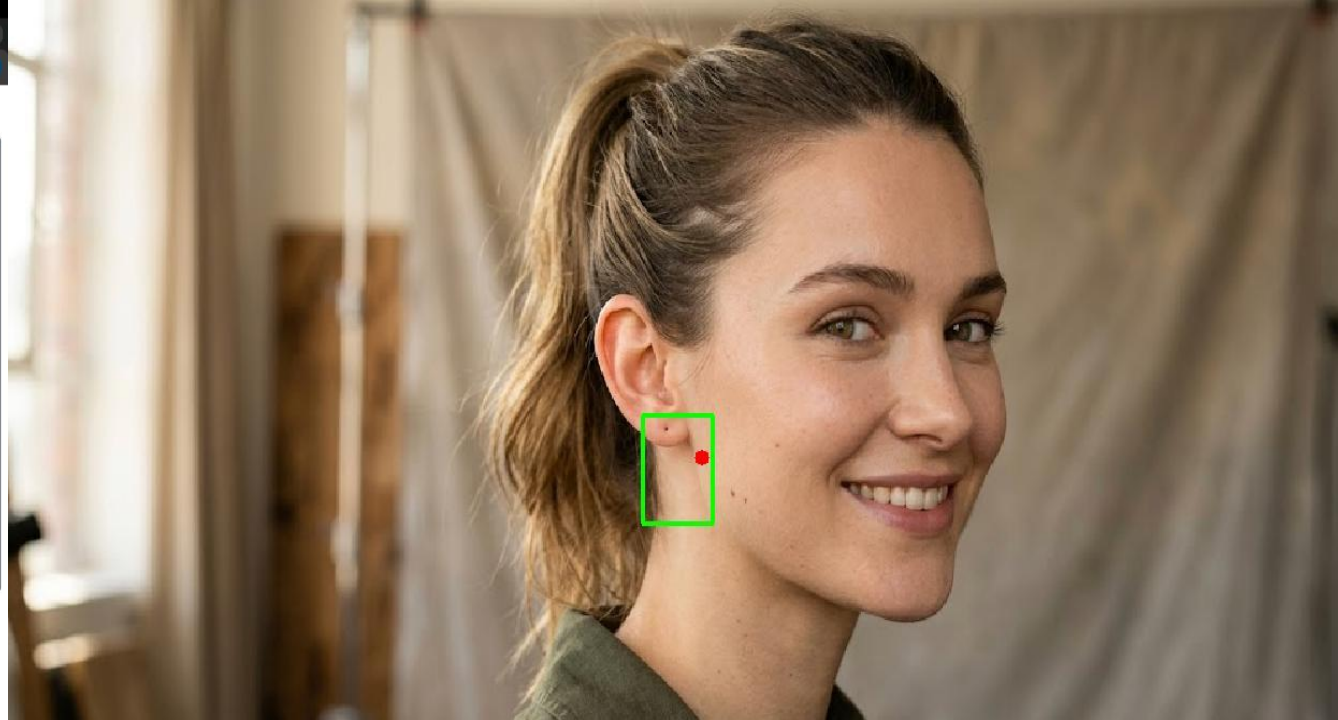
Raw Mask vs. Feathered Mask

0 200 400 600 800 1000 1200 1400 1600 1800 2000

0 500 1000 1500 2000

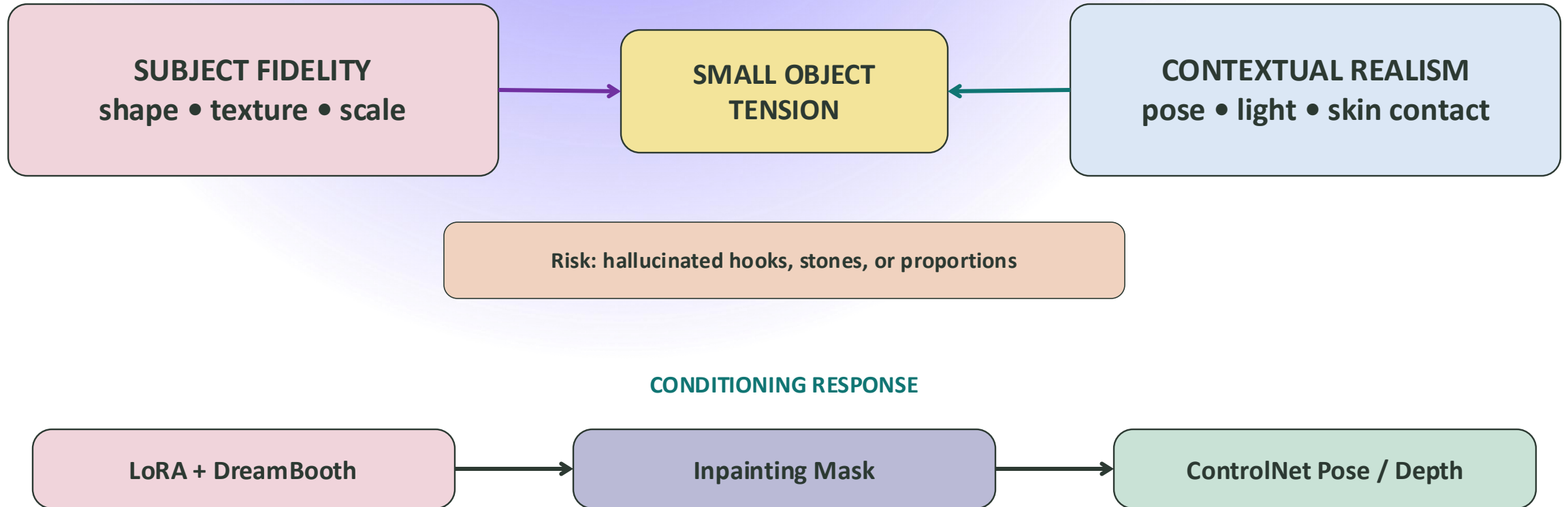
Table of contents

No sections detected



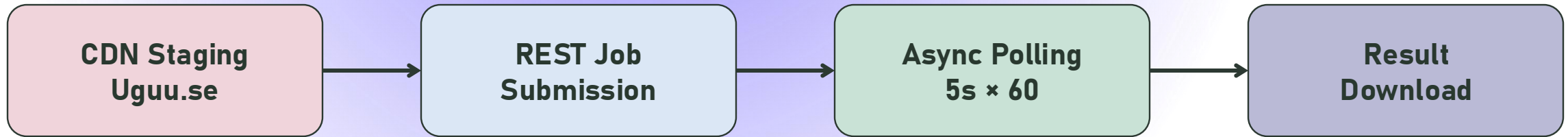
Fidelity vs. Contextual Realism

Small jewelry details are easy for diffusion models to reinterpret



Strategy B: Production API Orchestration

External inference, coordinated as a secure, credit-aware asynchronous job



pending → processing → succeeded

MLaaS

Claid.ai Fashion Models API

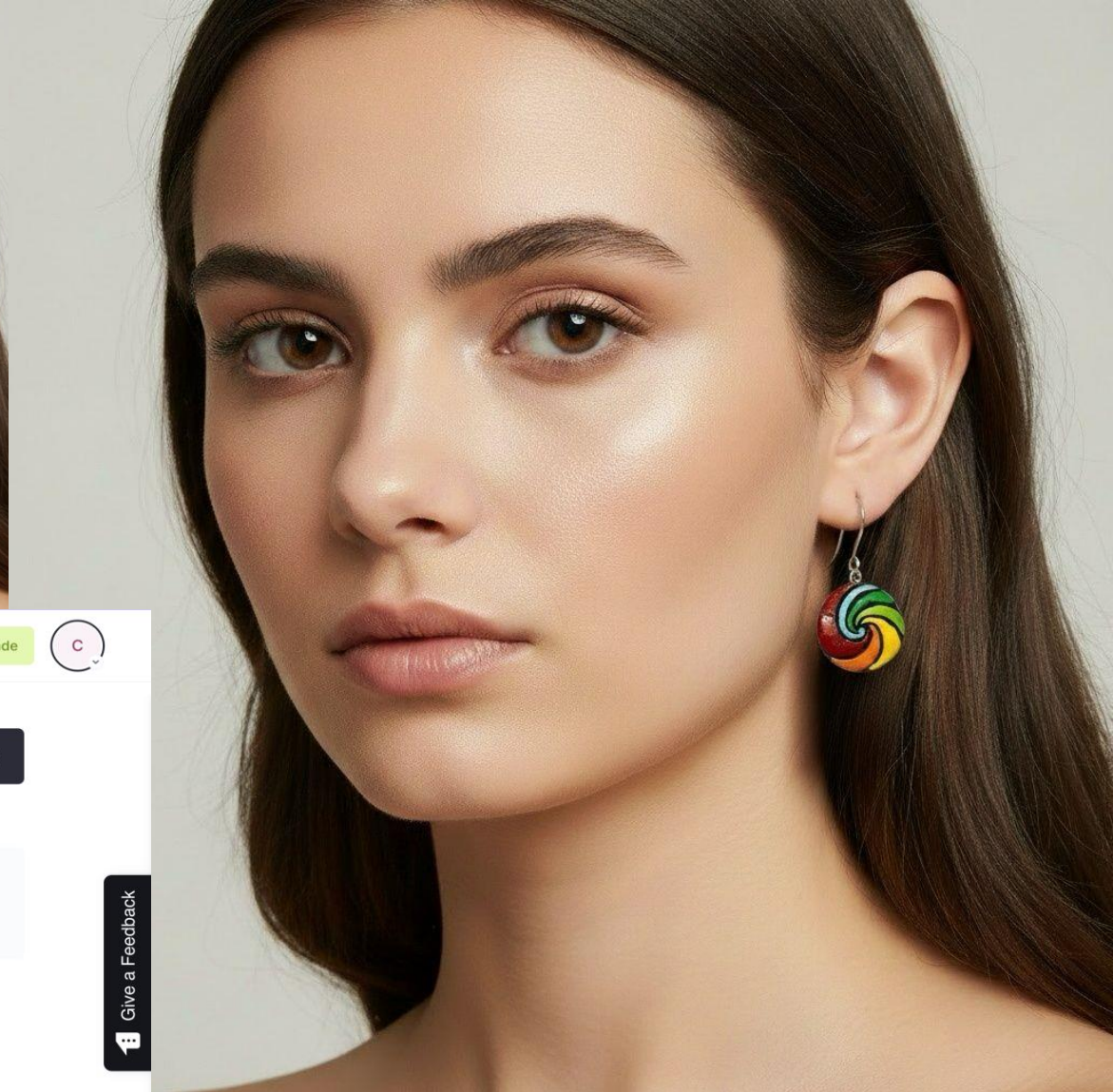
Production inference endpoint

Prompt Engineering for Constrained Generation

▪ Ear lobe-relative scale ▪ single-ear visibility ▪ shape and texture preservation

Secure API Key Orchestration | Local JSON Database ▪ Credit Tracking ▪ Dynamic Key Rotation

Prevents hardcoded credential leaks and automatically blocks execution when API credits are exhausted.



CLAID. AI by Let's Entrance

Help us make Claid API better

API Credits 0 / 0

Upgrade



Dashboard

Editing Studio

Integrations

Get Started

API Keys

Storage

Usage

Webhook settings

Documentation

API Playground

Account Settings

Key list

Create, manage and delete your API keys.

Create API key

earring_mockup_gen

7895f8*****

Access permissions: storage, image_editing, image_generation | Created on July 12, 2026

...

Give a Feedback



Two Tracks, One System

Production now, deeper control next

STRATEGY A / DIFFUSION

Strategy A (Diffusion) represents the long-term R&D direction: a fully self-hosted, infinitely customisable pipeline with no API dependency or credit cost.

The technical groundwork we laid with LoRA and DreamBooth is the foundation for the next iteration of this system.

STRATEGY B / API

Strategy B (API) gave us production-ready, consistent results immediately.

Speed to ship

Immediate reliability + long-term ownership

Business Impact & Conclusion

The Cost Impact

Max Cost Reduction

Eliminates the need for physical studios, professional lighting rigs, and paid fashion models. Marginal cost per mockup drops to zero.

The Time Impact

3 Weeks → 30 Seconds

Replaces a multi-week photography, logistics, and editing pipeline with a single-click, automated batch process.

The Scale Impact

Unlimited Scalability

Containerized Docker architecture guarantees absolute visual consistency, whether processing 50 SKUs or 5,000 SKUs.

The AkshayKala AI Studio transforms a logistical bottleneck into a scalable, zero-cost software pipeline.

Demo Video :

Google Drive : [Click here to view the demo](#)

Thank You